SMOOTHNESS OF A FINITE TYPE MAP EQUALS FLATNESS WITH PERFECT DIAGONAL; FOUR-WAY EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18528-18540 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18542-18556, the complete author proof (13 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Noetherian finite type hypothesis retained exactly; the perfect-diagonal input is a deep theorem cited, not restated.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms-chapter flatness of smooth maps and base change of smooth maps; Divisors, lemma section-smooth-regular-immersion; the chapter's regular-immersion-lci and lci-properties lemmas; Divided Power Algebra, lemma perfect-diagonal (a perfect diagonal plus flatness forces smoothness in the Noetherian finite type setting).

Source reference. Original lemma 067R: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18556–18556. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FCX: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18556–18556. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069H: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18150–18157. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 069L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18306–18310. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VB: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18556–18556. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 01VF: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18556–18556. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a finite type morphism of locally Noetherian schemes. Denote $\Delta : X \to X \times_Y X$ the diagonal morphism. The following are equivalent*

(1) *f is smooth,*
(2) *f is flat and $\Delta : X \to X \times_Y X$ is a regular immersion,*
(3) *f is flat and $\Delta : X \to X \times_Y X$ is a local complete intersection morphism,*
(4) *f is flat and $\Delta : X \to X \times_Y X$ is perfect.*

Proof. Assume (1). Then f is flat by Morphisms, Lemma 01VF. The projections $X \times_Y X \to X$ are smooth by Morphisms, Lemma 01VB. Hence the diagonal is a section to a smooth morphism and hence a regular immersion, see Divisors, Lemma 067R. Hence (1) $\Rightarrow$ (2). The implication (2) $\Rightarrow$ (3) is Lemma 069L. The implication (3) $\Rightarrow$ (4) is Lemma 069H. The interesting implication (4) $\Rightarrow$ (1) follows immediately from Divided Power Algebra, Lemma 0FCX. $\square$